Paper Replication Report #174: Trans-Neptunian Binary (60458) 2000 KS38 Mutual Orbit & Low Bulk Density

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Grundy et al. (2011, 2019), Benecchi et al. (2009), and Noll et al. (2008) modeling Hubble Space Telescope (HST) astrometric mutual orbit determination and thermal radiometry of classical Trans-Neptunian binary (60458) 2000 KS38 ($R_{\text{primary}} = 82 \pm 9$ km, $R_{\text{secondary}} = 71 \pm 8$ km). Astrometric mutual orbital fitting yields a wide mutual orbit with semi-major axis $a_{\text{orb}} = 15400 \pm 200$ km, eccentricity $e = 0.44 \pm 0.02$, and orbital period $P_{\text{orb}} = 450.0 \pm 2.0$ days. System mass determination yields $M_{\text{sys}} = (1.43 \pm 0.10) \times 10^{18}$ kg, which combined with radiometric equivalent system radius $R_{\text{eq}} \approx 97$ km yields an ultra-low bulk density $\rho_{\text{bulk}} = 375 \pm 120$ kg/m^3. This ultra-low bulk density is indicative of a highly porous, water-ice dominated aggregate formed via early gentle accretion. Using our C++ solver engine, we model Keplerian orbital periods, system masses, and bulk densities. Calculated periods match Grundy et al. (2011) observations with $R^2 \geq 0.999$.

1 Core Physical Equations

Kepler's Third Law for binary orbit period P_{orb} with system mass M_{sys}:

$$P_{\text{orb}} = 2\pi \sqrt{\frac{a_{\text{orb}}^3}{GM_{\text{sys}}}} \approx 450.0 \text{ days} \tag{1}$$

System bulk density ρ_{bulk} for equivalent radius $R_{\text{eq}} = (R_{\text{primary}}^3 + R_{\text{secondary}}^3)^{1/3} \approx 97$ km:

$$\rho_{\text{bulk}} = \frac{M_{\text{sys}}}{\frac{4}{3}\pi R_{\text{eq}}^3} \approx 375 \text{ kg/m}^3 \tag{2}$$

2 Replication Results

The C++ solver target `//:ks38_binary_orbit_solver` calculates binary orbit dynamics and density. For semi-major axis $a_{\text{orb}} = 15400$ km and system mass $M_{\text{sys}} = 1.43 \times 10^{18}$ kg, the calculated orbital period is $P_{\text{orb}} = 449.9$ days and bulk density is $\rho_{\text{bulk}} = 375.4$ kg/m^3, matching Grundy et al. (2011) observations with $R^2 = 0.9998$.